

FIFTH SEMESTER U.G. DEGREE EXAMINATION, NOVEMBER 2021

(CBCSS—UG)

Mathematics

MTS 5B 08—LINEAR PROGRAMMING

(2019 Admissions)

Time : Two Hours

Maximum : 60 Marks

Section A*Answer at least eight questions.**Each question carries 3 marks.**All questions can be attended.**Overall Ceiling 24.*

1. Define canonical minimization linear programming problem.
2. Give an example of a bounded polyhedral convex subset in $\mathbb{R}^2$.
3. State the canonical minimization linear programming problem represented by the following tableau :

x	1	2	3
y	4	5	6
-1	7	8	9

$= t_1 \quad = t_2 \quad g$

4. Define unbounded linear programming problem.
5. Pivot on 5 in the canonical maximum tableau given below :

x_1	x_2	-1	
1	2	3	$= -t_1$
4	5	6	$= -t_2$
7	8	9	$= f$

6. Write the simplex algorithm for maximum tableaux.
7. What do you mean by complementary slackness ?
8. State Duality theorem.

Turn over

9. Consider the canonical maximization linear programming problem given below ;

Maximize $f(x_1, x_2) = x_1$ subject to

$$x_1 + x_2 \leq 1$$

$$x_1 - x_2 \geq 1$$

$$x_2 - 2x_1 \geq 1$$

$$x_1, x_2 \geq 0$$

state the dual canonical minimization of the linear programming problem.

10. Distinguish between balanced and unbalanced transportation problem.
11. Using VAM to obtain a basic feasible solution of the transportation problem given below :

4	5	5
3	2	7
6	3	9
7	5	4
14	11	

12. Explain the minimum entry method for obtaining initial basic feasible solution in transportation problem.

(8 × 3 = 24 marks)

Section B

Answer at least five questions.

Each question carries 5 marks.

All questions can be attended.

Overall Ceiling 25.

13. Solve the following linear programming problem by geometrical method.

Maximize $f(x, y) = -2y - x$ subject to

$$2x - y \geq -1$$

$$3y - x \leq 8$$

$$x, y \geq 0.$$

14. Solve the following canonical linear programming problem using simplex algorithm :

x_1	x_2	-1	
-1	1	1	$= -t_1$
1	-1	3	$= -t_2$
1	2	0	$= f$

15. Solve the canonical linear programming problem using simplex algorithm :

x	-2	1	-3
y	1	-2	-2
-1	1	0	0

$= t_1 \quad = t_2 \quad g$

16. Solve the non-canonical linear programming problem given below

Maximize $f(x, y, z) = 2x + y - 2z$ subject to

$$x + y + z \leq 1$$

$$y + 4z = 2$$

$$x, y, z \geq 0.$$

17. Write the dual simplex algorithm for minimum tableaus.

18. Solve the transportation problem given below :

	M ₁	M ₂	M ₃	
W ₁	2	1	2	50
W ₂	9	4	7	70
W ₃	1	2	9	20
	40	50	20	

19. Apply Northwest-corner method to obtain the initial basic feasible solution of the transportation problem given below :

7	2	4	10
10	5	9	20
7	3	5	30
20	10	30	

(5 × 5 = 25 marks)

Section C

Answer any **one** question.
The question carries 11 marks.

20. Solve the canonical linear programming problem given below using the simplex algorithm.

x	y	z	-1	
1	2	1	4	$= -t_1$
2	1	5	5	$= -t_2$
3	2	0	6	$= -t_3$
1	2	3	0	$= f$

Turn over

21. Write the Hungarian algorithm. Using this algorithm solve the following assignment problem :

2	3	2	4
5	8	4	3
5	9	5	2
7	6	7	4

(1 × 11 = 11 marks)

CHMK LIBRARY UNIVRSITY OF CALICUT

FIFTH SEMESTER U.G. DEGREE EXAMINATION, NOVEMBER 2021**(CBCSS—UG)****Mathematics****MTS 5B 07—NUMERICAL ANALYSIS****(2019 Admissions)****(Multiple Choice Questions for SDE Candidates)****Time : 15 Minutes****Total No. of Questions : 15****Maximum : 15 Marks****INSTRUCTIONS TO THE CANDIDATE**

1. This Question Paper carries Multiple Choice Questions from 1 to 15.
2. The candidate should check that the question paper supplied to him/her contains all the 15 questions in serial order.
3. Each question is provided with choices (A), (B), (C) and (D) having one correct answer. Choose the correct answer and enter it in the main answer-book.
4. The MCQ question paper will be supplied after the completion of the descriptive examination.

MTS 5B 07—NUMERICAL ANALYSIS

(Multiple Choice Questions for SDE Candidates)

1. The floating-point form of π using five-digit chopping is :
(A) 3.1416. (B) 3.1415.
(C) 3.14159. (D) 0.3141.
2. If t is the largest non-negative integer for which $\frac{|p - p^*|}{|p|} \leq 5 \times 10^{-t}$, then t is called :
(A) Error bound. (B) Five-digit chopping.
(C) Significant digits. (D) None of these.
3. For $x^3 - 7x^2 + 14x - 6 = 0$ on $[0, 1]$, using Bisection method, p_3 lies in the interval :
(A) (0.5, 0.75). (B) (0.75, 1).
(C) (0.5, 0.625). (D) None of these.
4. Find the first approximation of $f(x) = x^3 + x - 1 = 0$ by Newton's method :
(A) 0.5. (B) 0.45.
(C) 0.24. (D) 0.75.
5. The linear Lagrange interpolating polynomial that passes through the points (2, 4) and (5, 1) is :
(A) $-x + 6$. (B) $-x + 5$.
(C) $x - 6$. (D) $x - 5$.
6. For $f(x) = \tan x$, and $x_0 = 0$, and $x_1 = 0.6$, find interpolation polynomial of degree at most one :
(A) $1.031121x + 1$. (B) $0.031121x - 1$.
(C) $1.031121x$. (D) $0.031121x - 1$.
7. The Trapezoidal rule is used for approximation of :
(A) Differentiation. (B) Integration.
(C) Both (A) and (B). (D) None of the above.

8. The three-point midpoint formula is used for the approximation of :
- (A) Differentiation. (B) Integration.
- (C) Both (A) and (B). (D) None of the above.
9. The approximation $\int_a^b f(x) dx \approx \frac{h}{2} [f(x_0) + f(x_1)]$ is the :
- (A) Newton's formula. (B) Simpson's 1/3rd rule.
- (C) Trapezoidal rule. (D) Simpson's 3/8th rule.
10. By Trapezoidal rule $\int_0^2 (x+1)^{-1} dx$ is :
- (A) 1.333. (B) 1.111.
- (C) 1.099. (D) 0.099.
11. By Simpson's rule $\int_0^2 e^x dx$ is :
- (A) 6.389. (B) 4.164.
- (C) 8.389. (D) 6.421.
12. What is the Lipschitz constant for the function $f(t, y) = \cos(yt)$ on $D = \{(t, y) \mid 0 \leq t \leq 1, -\infty < y < \infty\}$?
- (A) 0. (B) 1.
- (C) 2. (D) 3.
13. What is the Lipschitz constant for the function $f(t, y) = ty$ on $D = \{(t, y) \mid 0 \leq t \leq 1, -\infty < y < \infty\}$?
- (A) 0. (B) 3.
- (C) 2. (D) 1.